

MURALI KRISHNAN THIRUMALAI

609-779-1612 | muralikrishnant@gmail.com | www.linkedin.com/in/tmuraliik/ | github.com/muraliikrishnant | East Windsor, NJ

EDUCATION

Rutgers University, New Jersey

Expected Graduation: May 2026

M.S. in Computer Engineering (GPA: 3.75/4)

- Cloud Computing and Big Data, Deep Learning, Networking Protocols
- Cybersecurity Certification – Issued by Rutgers University

B.S. in Electrical and Computer Engineering - Cum Laude Honors

May 2025

- Honors: Rutgers University Dean's List – 3x

May 2024 – May 2025

SKILLS

Languages: Python, JavaScript, C++, HTML, CSS, React

Libraries: Pandas, NumPy, Pytorch, RestAPI, PyTest

CI/CD - Databases: Git, GitHub, PostgreSQL, MySQL, Firebase

Containerization/Cloud: Docker, Kubernetes, Linux/Bash Commands, Azure, Cloudflare

General Tools: Jira, Kanban

AI TOOLS: Cursor, Cody, Windsurf, GitHub Copilot, Claude, Codex

CYBERSECURITY TOOLS: Wireshark, Kali Linux, SIEM (ELK), Ettercap

EXPERIENCE

TARS Group – AI Marketing & Automation Intern – Software Engineering Role

Sep 2025-Dec 2025

- Built a full-stack AI newsletter generation system using Python (FastAPI), Gemini/Ollama LLMs, and Tavily web research, featuring a React-style frontend with Human-in-the-Loop (HITL) review workflow, enabling iterative feedback and revision cycles before triggering final SMTP delivery, deployed on Render (backend) and GitHub Pages (frontend).
- Built and iterated a FastAPI–React AI chatbot integrating Google Gemini, implementing intent detection, FAQ-based retrieval, safety guardrails, and WordPress-embeddable UI to improve user engagement and reduce manual support load.
- Developed a real-time autonomous-vehicle deployment map using Firebase Realtime Database, Leaflet.js, and Firebase Hosting, including data schema design, live update listeners, and interactive geospatial visualizations for operational insights.
- Collaborated with supervisors in an agile environment, using tools like JIRA and Git to manage multi-week deliverables, perform testing and debugging, add logging and quality controls, and document architectures and incident-style findings to support informed decision-making.

PROJECTS

Attention Aware Mirror System – Mobile App

Sep 2024 - May 2025

- Cross-Platform App Development: Built a React Native application for iOS and Android using Expo, supporting secure user login and command handling.
- Integrated Firebase Authentication and Google OAuth to ensure secure session management and identity verification.
- Implemented low-latency TCP socket communication between the mobile client and Raspberry Pi 5 for hardware control.
- Synchronized mirror configurations using Firebase Realtime Database with strict security rules for authenticated R/W access.

Anzom Website Development – HTML | CSS | Javascript

Jan 2024 - May 2024

- Implemented seller product-creation flow with client-side validation and JSON form submission to persist new items via the /api/product endpoint into PostgreSQL.
- Wired category, subcategory, product, and featured-product GET APIs to the landing-page frontend so users can browse categories, product lists, and featured items from live data.

- Linked seller inventory to the main catalog by attaching new products to the correct category, subcategory, and seller ID, making them immediately searchable and visible on the landing page.
- Assisted with designing and implementing product-view APIs (by category, subcategory, and product ID) with consistent JSON responses and SQL queries powering catalog and seller views.

Man-in-the-Middle Attacks via ARP Spoofing on Public Wi-Fi Networks

Nov 2024 - Dec 2024

- Executed controlled ARP spoofing campaigns in lab Wi-Fi networks to study L2/L3 behavior and plaintext HTTP exposure, using Wireshark/Ettercap to capture and analyze traffic flows.
- Executed controlled ARP spoofing campaigns in lab Wi-Fi networks to study L2/L3 behavior and plaintext HTTP exposure, using Wireshark/Ettercap to capture and analyze traffic flows.
- Tested SSLStrip and HTTPS downgrade attempts, noting limitations imposed by HSTS and modern browser protections.
- Evaluated network-layer mitigations (static ARP, VPN, HTTPS enforcement) and documented trade-offs as part of an IEEE-style technical report focused on securing production Wi-Fi environments.

HACKATHON

Multi-Domain Support Triage Agent — HackerRank Orchestrate Hackathon

May 2026

- Built a terminal-based support triage agent in Python capable of classifying, retrieving, and responding to tickets across three distinct domains (HackerRank, Claude, Visa), processing 29 real-world tickets with domain-aware routing and grounded response generation.
- Designed a modular RAG pipeline with a custom retriever over a 2,659-chunk corpus scraped from live support sites, with MMR-based re-ranking to reduce redundant context and improve answer grounding.
- Implemented a multi-layer safety and escalation system covering fraud detection, prompt-injection attempts, dangerous instruction patterns, billing disputes, and account access, using both rule-based keyword matching and LLM-backed confidence scoring to prevent hallucinated or contradictory responses.
- Integrated an optional LLM response generation backend (Gemini 2.5 Flash) with post-generation contradiction guards, secure .env-based credential handling, and deterministic offline fallback, producing a full output.csv and human-readable log.txt transcript for judge evaluation.

CERTIFICATIONS

Python (Fundamentals & OOP), HackerRank (2024)

SQL (Querying & Relational Databases), August 2024, LinkedIn Learning (2024)

CC (ISC2) – In Progress 2026

Cybersecurity Certification – Issued by Rutgers University